

Learning Outcomes

By the end of this lab session, students will be able to

- Handle missing values appropriately.
- Encode categorical variables.
- Scale numerical features.
- Build preprocessing workflows using Column Transformer.
- Create reusable preprocessing Pipelines.
- Prevent data leakage during preprocessing.

Dataset:

- Titanic Dataset

Part A – Data Cleaning

1. Import the dataset by using,

```
import seaborn as sns  
df = sns.load_dataset("titanic")
```

2. Identify the feature types in the dataset.
 - Numerical features
 - Categorical features
 - Target variable

3. Check the dataset for
 - Missing values
 - Duplicate records

Remove duplicate records if any.

4. Identify the features containing missing values. Choose and justify an appropriate imputation method for each feature. Display the number of missing values before and after imputation.

Part B – Categorical Data Encoding

1. Identify,
 - Binary categorical features
 - Nominal categorical features
 - Ordinal categorical features
2. Apply and justify the most appropriate encoding technique for each categorical feature. Display the transformed dataset.

- Binary Encoding / Label Encoding
- One-Hot Encoding
- Ordinal Encoding (if applicable)

Part C – Feature Scaling

1. Apply the following scaling techniques to the numerical features and display the transformed values.
 - **StandardScaler**
 - **MinMaxScaler**
 - **RobustScaler**
2. Compare the outputs.
 - Which scaler produces values between 0 and 1?
 - Which scaler produces a mean close to 0?
 - Which scaling method would you recommend for this dataset? Explain.

Part D – Feature transformation

1. Identify the skewed numerical features using the histograms.
2. Apply an appropriate transformation to each skewed numerical feature and justify your choice.
3. Compare the distributions before and after transformation.

Part E – Case Study

A data analyst is preparing the Titanic dataset for a machine learning model. Before splitting the dataset into training and testing sets, the analyst performs missing value imputation and feature scaling using the entire dataset.

Another analyst follows the recommended workflow by first splitting the dataset into training and testing sets, then fitting the preprocessing methods using only the training dataset before transforming both datasets.

Answer the following questions:

1. What is the main difference between the two preprocessing workflows?
2. Which workflow introduces data leakage?
3. Why preprocessing methods should be fitted only on the training dataset?
4. Which workflow would you recommend for building a reliable machine learning model? Justify your answer.